Selvaragavan.M

AI Developer / ML Engineer

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Portfolio:SELVARAGAVAN.DEV

PROFILE:

AI & Data Science student and practicing Robotics Engineer who has developed LLM-powered agentic systems, designed NLP pipelines, and implemented production-style ML environments for real-world applications. Proficient in Python, PyTorch, Hugging Face Transformers, LangChain, and FastAPI. Engineered and deployed real-world AI solutions utilizing BERT, Mistral 7B, and DistilBERT fine-tuning, RL-based simulation, and multi-model ensemble architectures. Seeking AI Developer / ML Engineer roles to develop innovative machine learning solutions that drive automation, enhance data-driven decision-making, and optimize business processes.

TECHNICAL SKILLS:

- Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS
- ML/Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, Transformers, Reinforcement Learning
- LLMs & Agentic AI: Hugging Face Transformers, BERT, DistilBERT, Mistral 7B, LangChain, LangGraph, Prompt Engineering, Fine-tuning (LoRA basics), RAG pipelines, Kubernetes, SAS Programming, DevOps
- NLP: Text classification, tokenization, sequence labeling, model evaluation (F1, precision, recall)
- Computer Vision: YOLOv8, OpenCV, real-time object detection, custom dataset training
- MLOps & DevOps: OpenEnv-compatible RL environments, CI/CD basics, GitHub Actions, Docker (basics)
- API & Deployment: FastAPI, REST APIs, Streamlit, Gradio, Vercel
- Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, feature engineering
- Cloud: AWS (EC2, S3, Lambda, Redshift, SageMaker basics) AWS-certified, Azure, Cloud infrastructure
- Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux CLI, version control (Git)
- Databases: MySQL, PostgreSQL, Vector DBs (ChromaDB basics)

SOFT SKILLS:

- Strong communication, cross-functional collaboration, problem-solving, ownership, Critical thinking, Teamwork and awareness of ethical and responsible AI practices.

EDUCATION:

B. Tech — Artificial Intelligence & Data Science
Kathir College of Engineering, Coimbatore (2023-2027 pursuing) CGPA: 7.8

Relevant Coursework: Machine Learning, Deep Learning, Neural Networks, Data Structures & Algorithms, Database Management, Computer Vision, Statistics & Probability

EXPERIENCE:

Robotic Engineer at Robential Kinetics pvt.ltd: (Dec 2025 - Present)

- Led end-to-end design, programming, and testing of robotic systems for automation projects, owning the process from concept to deployment.
- Integrated hardware, sensors, and control software to enhance the reliability and efficiency of robotic arms and custom mechanisms.

WORKSHOPS & TRAINING:

- Completed an offline Data Structures and Algorithms workshop by LeSuccess,
- Attended a Data Science workshop by Amypo Technologies, Coimbatore
- Completed a one-month Virtual Internship Training on AWS Cloud Computing by NASSCOM FutureSkills Prime & SmartBridge.

PROJECTS:

Agentic Pentesting CLI using Custom LLM:

- Built a CLI-based agentic pentesting tool that uses a custom fine-tuned model to plan and execute penetration testing steps autonomously.
  GitHub: https://github.com/jas0nt0dd/pentest-ai-

Real-Time Object Detection using YOLO:

- Implemented a computer vision pipeline using YOLO for real-time object detection on images/video streams with Python and OpenCV
  GitHub:https://github.com/robentialkinetics-kishtech/zkbot-autonomous-cup-washing-system-

Policylens AI- AI Powered Policy Bias Detection & SDG Compliance Tool

- Built a full-stack AI tool that parses government policy PDFs and detects **5 dimensions of bias** — Gender, Socioeconomic, Procedural Fairness, Regional, and Ethnic/Caste — at clause level using a 3-model ensemble (BERT + IBM AIF360 + Mistral 7B
  GitHub:https://github.com/jas0nt0dd/policylens-ai

MLOps Incident Response Environment

- Developed an OpenEnv-compatible **MLOps incident response simulator** with task-based RL evaluation, component-level observations, and automated diagnosis scoring for production-style ML failures.
  GitHub: https://github.com/jas0nt0dd/mlops-incident-env

AWARDS & ACHIEVEMENTS:

- Winner, Eureka Bug Bounty Competition, IIT (BHU) – secured first place for identifying impactful security vulnerabilities in the target application.
- Shortlisted for the final round (Top 20 teams) in the AWS Campus Hackathon (Feb 2026) at IIT Kanpur, for building an AWS-based solution during Techkriti'26.
- **Finalist- AI Tool Development Challenge 2.0** (April 2026)
  Built PolicyLens AI, an open-source civic-tech tool for AI-driven bias detection in government policies; presented to judges in a competitive Zoom evaluation round; recognized for innovation in SDG 16 — Peace, Justice & Strong Institutions
- Placed in Top 30 on the leaderboard in **Vista – Codefest'26 Computer Vision Hackathon (Kaggle, IIT (BHU) Varanasi)**.
- Placed in Top 30 in **Enigma – Codefest Kaggle Competition** focused on algorithmic/ML problem-solving.

COURSES AND CERTIFICATIONS:

- AWS Cloud Computing Virtual Internship - NASSCOM FutureSkills Prime & SmartBridge
- Data Structures & Algorithms - LeSuccess (Offline Workshop)
- Data Science - Amypo Technologies, Coimbatore
- Generative AI for Data Analytics - Udemy
- AWS SimuLearn: Generative AI Practitioner - AWS Training & Certification
- Amazon Redshift Getting Started - AWS Training & Certification
- AWS SimuLearn: Cloud Computing Essentials - AWS Training & Certification
- Prompt Engineering – Udemy
- Ignite Bootcamp: Venture Idea Development - Wadhwani Foundation
- Project Horizon: The GPAi Case Competition - IIT Madras

Drive link: DRIVE

LANGUAGE: English, Tamil